Suyog Mainali

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Professional Summary

Senior Data Engineer with 7+ years of experience designing scalable data pipelines, ETL workflows, and cloud data platforms using Spark, SQL, and AWS. Experienced in building reliable, production-grade data systems for analytics and ML workloads.

Core Skills

Data Engineering: Python, SQL, Spark (PySpark), Spark SQL, ETL/ELT pipelines, Airflow, dbt, Kafka
Cloud & Warehousing: AWS (EMR, S3, Glue, Lambda), GCP (BigQuery, GCS, Dataproc), Snowflake, Redshift
Data Architecture: data pipelines, schema design, dimensional modeling, data marts, data governance, data quality
Databases: PostgreSQL, MySQL, MongoDB, Elasticsearch
Distributed Systems: Hadoop (HDFS, Hive), Delta Lake, Parquet, JSON, query optimization
Platform & Ops: Docker, Kubernetes, CI/CD, Git, Linux, SLAs/SLOs, monitoring, alerting, incident RCA
ML & Data Quality: feature engineering, model-ready datasets, data validation, schema enforcement
Analytics & BI: Tableau, Power BI, metrics reporting, dashboards

Professional Experience

Zakipoint Health *Senior Data Engineer | July 2020 – July 2025*

- Built real-time cost-estimation data services (Java/Spring Boot REST APIs) backed by Python/Spark/SQL pipelines processing 10TB+ daily, delivering sub-second pre-visit cost estimates.
- Re-architected distributed PySpark and SQL ETL pipelines on AWS Glue and EMR with automated data validation and observability, reducing failures by 60%+ and improving performance through partitioning, caching, and optimization.
- Owned dataset SLAs end-to-end (SLIs/SLOs, monitoring, alerting, incident RCA), maintaining 99.8% uptime in a HIPAA-compliant environment.
- Developed analytics-ready datasets and data models using Spark SQL and Hive transformations, enabling near real-time BI dashboards and reducing reporting latency from 2 days to minutes.
- Designed dimensional models (star schema, fact/dimension tables, SCD) and curated data marts to standardize metrics for analytics, experimentation, and reporting.
- Built production ML feature pipelines supporting risk scoring and care-gap detection models, reducing data preparation time by ~40% and improving feature consistency and model reliability.
- Collaborated with product, analytics, and engineering teams to translate business requirements into scalable data pipelines and reusable datasets.

Om Hospital *Data Engineer | April 2018 – July 2020*

- Led migration from legacy and paper-based systems to a digital health platform supporting 250K+ annual patient visits, enabling centralized analytics and reporting.
- Built the organization's first centralized data warehouse integrating clinical, operational, and laboratory data, establishing a single source of truth.
- Developed Python-based ETL pipelines and data workflows supporting 10K+ monthly patient interactions and longitudinal data tracking.
- Built clean, structured datasets to support early-stage ML use cases for patient risk analysis and operational reporting.
- Trained 50+ staff on digital data workflows, improving data quality, adoption, and reporting consistency.

Projects

Healthcare Churn & Specialty Analytics (HIPAA-Restricted)
Built end-to-end analytics and ML pipelines on healthcare data; engineered features, prepared modeling datasets, and developed predictive models for churn and specialty utilization risk insights.
Technologies: R, R Markdown, tidyverse, caret, randomForest, xgboost, logistic regression, LDA, ggplot2, Python, pandas

Amazon Reviews ETL Pipeline GitHub Repository
Built a production ETL pipeline to extract and process product metadata and review data, creating validated PostgreSQL datasets for downstream analytics and ML-ready use cases.
Technologies: Python, Flask, PostgreSQL, SerpAPI, BeautifulSoup, ETL, REST API, HTML, CSV export, data validation

Professional Certifications

Mastering Azure Databricks for Data Engineers – Packt/Coursera (Certificate)
Google Data Analytics – Google/Coursera (Certificate)
Data Analysis with R – Duke University/Coursera (Certificate)

Education

William & Mary, Mason School of Business | MS Business Analytics — Expected May 2026
Vellore Institute of Technology | B.Tech Computer Science Engineering — 2018 – 2022

Work Authorization: F-1 STEM OPT (3 years U.S. work eligibility; sponsorship required thereafter)